Problem 03: Gossip Chain

Time limit: 5 seconds

In a small town, gossip spreads fast. Whenever one friend talks to another, gossip travels along that direction. Each record of gossip is written as: $u\ v\ t$ meaning that friend u told friend v something at time t.

Over time, the same people may exchange information many times in different directions. We would like to understand *how gossip typically flows among groups of three friends* within a short period of time.

Every gossip exchange is a directed message **(u → v)** occurring at a certain time. We are interested in three-message gossip bursts involving exactly three distinct friends **(A, B, C)**, with strictly increasing times ***t1 < t2 < t3***, and where all three messages happen within **Δ minutes**, i.e. **t3 - t1 ≤ Δ**.

The following three gossip patterns are of interest:

1. **A → B, B → C, C → A**
2. **A → B, A → C, B → C**
3. **A → B, C → B, B → A**

Given all gossip events recorded for a single day, count how many distinct three-message sequences correspond to **Type 1, Type 2, and Type 3** gossip patterns.

Input:

The first line contains three integers:

N M Δ

- **N (1 ≤ N ≤ 5000):** number of friends
- **M (1 ≤ M ≤ 50000):** number of gossip events
- **Δ (1 ≤ Δ ≤ 5000):** maximum time window (in minutes)

Each of the next **M** lines contains three space-separated integers: $u\ v\ t$, indicating that friend **u** told friend **v** something at time **t (0 ≤ u, v < N, u ≠ v, 1 ≤ t ≤ 10^9)**. All gossip events are distinct.

Output:

Print three integers separated by spaces:

CountType1 CountType2 CountType3

Sample Input	Sample Output
3 12 2 0 1 1 1 2 2 2 0 3 0 1 6 0 2 7 1 2 8 0 1 11 2 1 12 1 0 13 0 1 16 1 2 17 2 0 18	2 1 1